Introduction to Quantum Optics

(10 marks)

Question 1: Two-Mode Squeezed Vacuum

Solution

(a) Reduced state of one mode

We trace over the idler mode to obtain the reduced density matrix for the signal mode:

$$\hat{\rho}_s = \mathrm{Tr}_i\left(|\Psi\rangle\langle\Psi|\right) = (1-\lambda^2)\sum_{n=0}^{\infty}\lambda^{2n}|n\rangle_s\langle n|.$$

This is the density operator of a thermal state with geometric photon-number distribution:

$$p_n = (1-\lambda^2)\lambda^{2n}.$$

The mean photon number is:

$$\bar{n} = \sum_{n=0}^{\infty} n p_n = \frac{\lambda^2}{1-\lambda^2} = \sinh^2 r.$$

(b) Cross-correlation function $g_{si}^{(2)}(0)$

For the given state, we have perfect photon-number correlations: $n_s = n_i = n$ for each term. Therefore:

$$\langle \hat{n}_s \hat{n}_i \rangle = \sum_{n=0}^{\infty} n^2 p_n.$$

Using the thermal distribution:

$$\langle n^2 \rangle = 2\bar{n}^2 + \bar{n},$$

so:

$$\langle \hat{n}_s \hat{n}_i \rangle = \langle n^2 \rangle = 2\bar{n}^2 + \bar{n}.$$

Since $\langle \hat{n}_s \rangle = \langle \hat{n}_i \rangle = \bar{n}$, we find:

$$g_{si}^{(2)}(0) = \frac{2\bar{n}^2 + \bar{n}}{\bar{n}^2} = 2 + \frac{1}{\bar{n}}.$$

(c) Heralded single-photon generation

For small squeezing ($\lambda \ll 1$), the state approximates:

$$|\Psi\rangle \approx |0,0\rangle + \lambda|1,1\rangle + O(\lambda^2).$$

Detecting one photon in the idler mode projects the signal mode onto:

$$|\Psi_{\text{heralded}}\rangle \approx |1\rangle_s,$$

up to normalization. Thus, in the low-gain limit, detection of an idler photon *heralds* a single-photon state in the signal mode.

Question 2: Beam splitter

(a) Beam splitter

$$|n\rangle_A |0\rangle_B = \frac{(\hat{a}_A^\dagger)^n}{\sqrt{n!}} |0\rangle_A |0\rangle_B \tag{1}$$

Invert BS transformation:

$$\begin{pmatrix} \hat{a}_A \\ \hat{a}_B \end{pmatrix} = B^\dagger \begin{pmatrix} \hat{a}_{A'} \\ \hat{a}_{B'} \end{pmatrix} = \begin{pmatrix} \tau & \varrho \\ -\varrho & \tau \end{pmatrix} \begin{pmatrix} \hat{a}_{A'} \\ \hat{a}_{B'} \end{pmatrix} \tag{2}$$

$$|n\rangle_A |0\rangle_B \rightarrow \frac{1}{\sqrt{n!}} (\tau \hat{a}_{A'}^\dagger + \varrho \hat{a}_{B'}^\dagger)^n |0\rangle |0\rangle \tag{3}$$

$$= \frac{1}{\sqrt{n!}} \sum_{k=0}^{n} \binom{n}{k} (\tau \hat{a}_{A'}^\dagger)^k (\varrho \hat{a}_{B'}^\dagger)^{n-k} |0\rangle |0\rangle \tag{4}$$

$$= \frac{1}{\sqrt{n!}} \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \tau^k \varrho^{n-k} \sqrt{k!} \sqrt{(n-k)!} |k\rangle_{A'} |n-k\rangle_{B'} \tag{5}$$

$$= \sum_{k=0}^{n} \sqrt{\frac{n!}{k!(n-k)!}} \tau^k \varrho^{n-k} |k\rangle_{A'} |n-k\rangle_{B'} \tag{6}$$

(b) Coherent state input

Recall that a coherent state is generated by a displacement:

$$|\alpha\rangle_a |0\rangle_b = \big(D_a(\alpha) \otimes \mathbb{I}_b\big) |0,0\rangle, \qquad D_a(\alpha) = \exp(\alpha \hat{a}^\dagger - \alpha^* \hat{a}).$$

Let U_{BS} be the beam splitter unitary with $U_{\mathrm{BS}}^\dagger \, \hat{a} \, U_{\mathrm{BS}} = \frac{\hat{c}+\hat{d}}{\sqrt{2}}$ and $U_{\mathrm{BS}}^\dagger \, \hat{b} \, U_{\mathrm{BS}} = \frac{\hat{c}-\hat{d}}{\sqrt{2}}$. Since $U_{\mathrm{BS}}|0,0\rangle = |0,0\rangle$,

$$U_{\mathrm{BS}}|\alpha,0\rangle = U_{\mathrm{BS}} \, D_a(\alpha) \, U_{\mathrm{BS}}^\dagger \, |0,0\rangle = \exp\Big(\alpha \, U_{\mathrm{BS}} \hat{a}^\dagger U_{\mathrm{BS}}^\dagger - \alpha^* \, U_{\mathrm{BS}} \hat{a} U_{\mathrm{BS}}^\dagger\Big) |0,0\rangle.$$

Using $U_{\mathrm{BS}} \hat{a}^\dagger U_{\mathrm{BS}}^\dagger = \frac{\hat{c}^\dagger + \hat{d}^\dagger}{\sqrt{2}}$ and the Baker–Campbell–Hausdorff factorization for displacements,

$$U_{\mathrm{BS}}|\alpha,0\rangle = D_c\Big(\tfrac{\alpha}{\sqrt{2}}\Big) \, D_d\Big(\tfrac{\alpha}{\sqrt{2}}\Big) \, |0,0\rangle = \Big|\tfrac{\alpha}{\sqrt{2}}\Big\rangle_c \otimes \Big|\tfrac{\alpha}{\sqrt{2}}\Big\rangle_d.$$

Hence the output is a *product of coherent states* with amplitudes $\alpha/\sqrt{2}$ in each arm.

(c) Photodetection statistics in mode c

Mode c is in a coherent state $\big|\alpha/\sqrt{2}\big\rangle$, so the detected photon-number distribution is Poissonian:

$$P(n) = e^{-|\alpha|^2/2} \, \frac{\big(|\alpha|^2/2\big)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

Thus, ideal photon counting registers Poissonian statistics with mean $|\alpha|^2/2$.